



Troubleshooting Bible (TSB)

Beta 1.0

Updated: 5/28/2021

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Innovation in technology...

Part of what makes SandboxVR so innovative is its ability to push the limits of technology and boundaries to create unique and unmatched experiences for our guests. A wand may just be a wand, but then we are challenged to engage our imaginations and see beyond the endless possibilities of what it could become. Every day we continue to take these steps toward creating new updates, upgrades, and bringing everything together to reject reality, and substitute our own. We strive to ensure each visit back to our Holodeck is unique and ever evolving.

When The Going Gets Tough...

As with any technology, unexpected things can occur, causing the equipment to not work properly. While other companies might panic or are ill-equipped to fix the issues in a timely manner for the guests...we are always prepared!

The following is a mixture of the most common and unique troubleshooting fixes. This is a living, breathing document, so expect frequent updates to keep our hard-working staff knowledgeable and prepared.

Finagle's Law:

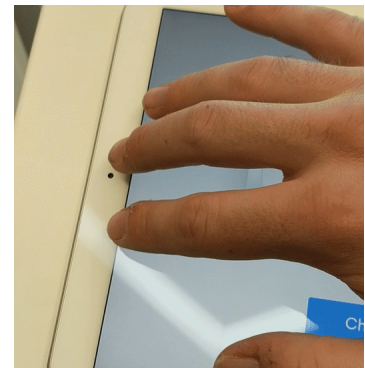
"Anything that can go wrong, will - at the worst possible moment."

1. Check-In

A. Check-In Tablets

a. No Bookings Shown at All

1. Check WiFi
 - If off, Slack for AP issues
2. Signout/Sign In
 - Drag 3 fingers across the screen
 - Select *Signout*
 - Go to *google.com/device*
 - Enter code
 - Select *Verify*
3. Try again
4. If problem remains, Slack for assistance and notify leads and managers



b. Booking for A Single Reservation Not Showing Up on Tablet

1. Check email on booking is real
 - If email is not real, update on Checkfront and repeat
 - If email is real, Slack & call for assistance

Reservations on 1 invoice, but for 2 different days = 2nd date will not be recognized by system.

Item
[4]Hillsdale Shopping Center Sat May 22, 2021 08:15 PM
[4]Hillsdale Shopping Center Sun May 30, 2021 03:15 PM

2. Experience

A. Experience Shortcuts

(during experience on Server 1)

Function	Keys	Notes
Toggle full-screen view	Alt + Enter	No button available on Console
Toggle Console	Shift + Space + T	
Toggle Log	Shift + Space + L	
Start Game	Shift + Space + S	
Pause / Resume Game	Shift + Space + P	
Force Defeat	Shift + Space + E	
Go to Wave	Shift + Space + Alpha #	Select the number of the wave from the alphanumeric row
Kill Player	Shift + Space + Keypad #	Select the number of the player from the keypad
Solve OilBarrelPuzzle	Shift + Space + Q	Only available for Davy Jones

B. Calibrating Cameras:

Video Instructions: [Calibrating Cameras](#)

(To be done every morning before room is in use and at manager's discretion during operating hours)

1. ENABLE Strobes
2. "Start Masking" for 20 seconds, then "Stop Masking"
3. DISABLE Strobes
4. "Start wave in camera" and waving the T-wand in front of each camera. (cameras need to be under 0.1 or very close to that range and in green). *If you see image error is greater than 0.175(see right), repeat step 4.*
6. ENABLE Strobes
7. Place the T-wand on the center floor (look for the cross) with the long end pointing at the fans, then turn on the wand
8. "Start Set Origin" for 10 seconds, then "Complete set origin"

Vicon Motion Capture System Calibration Guide



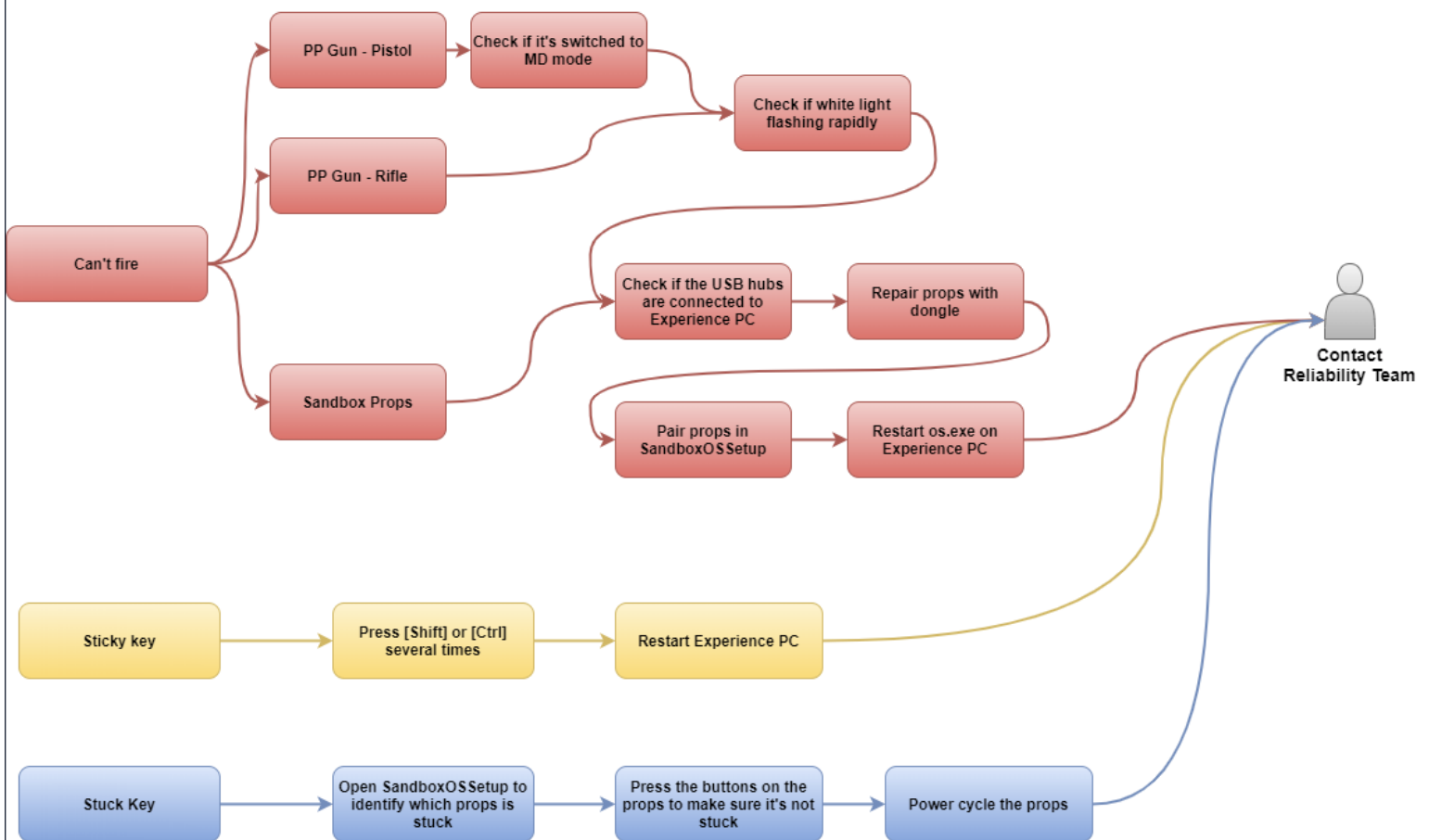
100%		
Camera	Wand Count	Image Error
1 Vero v2.2	3150	4.538
2 Vero v2.2	1405	6.611
3 Vero v2.2	1013	3.259
4 Vero v2.2	4371	2.436
5 Vero v2.2	4436	1.563
6 Vero v2.2	8080	1.567
7 Vero v2.2	1994	1.512
8 Vero v2.2	2147	1.630
9 Vero v2.2	1000	1.960
10 Vero v2.2	1316	1.590
11 Vero v2.2	3194	1.852
12 Vero v2.2	4949	4.311

C. Quick Troubleshooting Guides:

ALWAYS PAUSE EXPERIENCE DURING SESSIONS! (Communicate to guest)

Props

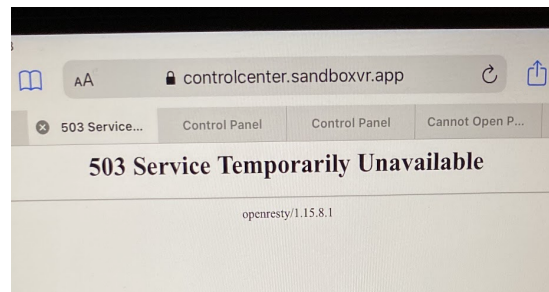
Props Troubleshooting



D. Experience Won't Launch on iPad:

When you see the error (on the right):

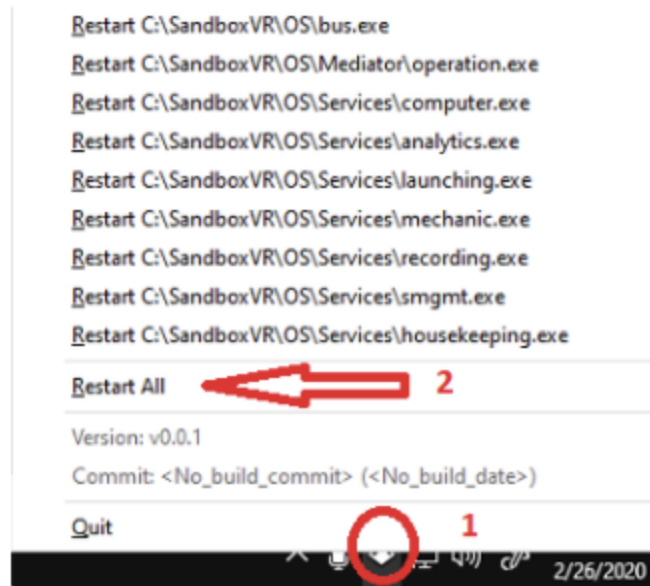
- Make sure iPad is connected to store WIFI network
- Restart Experience server
- If problem remains, Slack for assistance and notify leads and managers



E. Guns / Pistols Not Working

Video Instructions: [malfunction props video](#)

- Check if it switched "OFF" or the gun's battery is out.
- Power cycle.
- Check pistols power switch in the center; rifles switcher on "MD"
- Reset the affected gun's dongle (*from USB Hubs for rifles and pistols; from USB Hubs AND prop on wands*)
- Restart SandboxVR OS
- Open SandboxVR OS Setup on server 1, to check if guns/pistols/wands are active - by pulling triggers on guns/pistols/wands to see if it shows greenlights on OS. If not, clear and repair. CLOSE the OS after pairing.
 - *Close SandboxOSSetup and select Restart All for changes to apply*



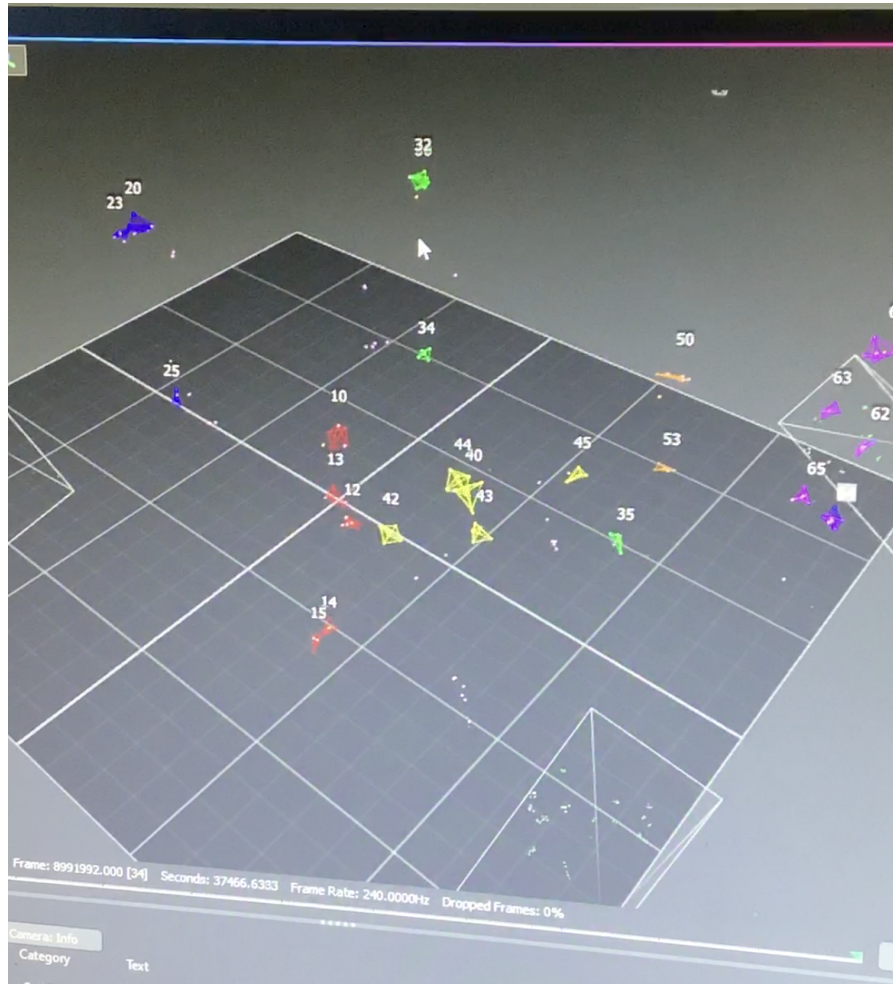
- If problem remains, Slack for assistance and notify leads and managers

F. Wands Not Working

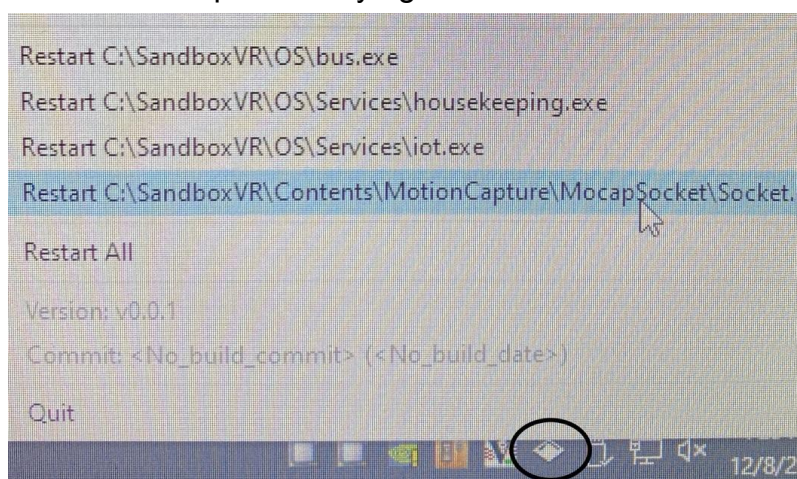
G. Tracking Issues & MOCAP

a. Motion Tracking issue:

- Switch to Server 2 to check on Evoke in 3D scene view to see if everything is tracking correctly. **Re-calibrate** room if you see a lot of **pink dots (noise)** in Evoke 3D scene (see below)



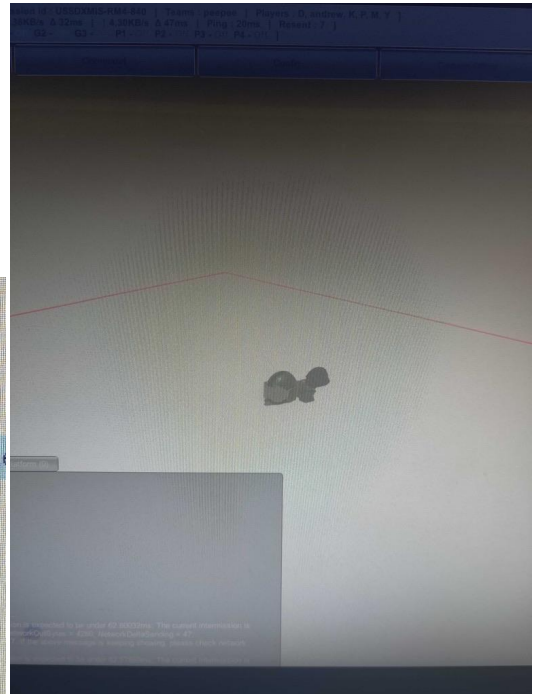
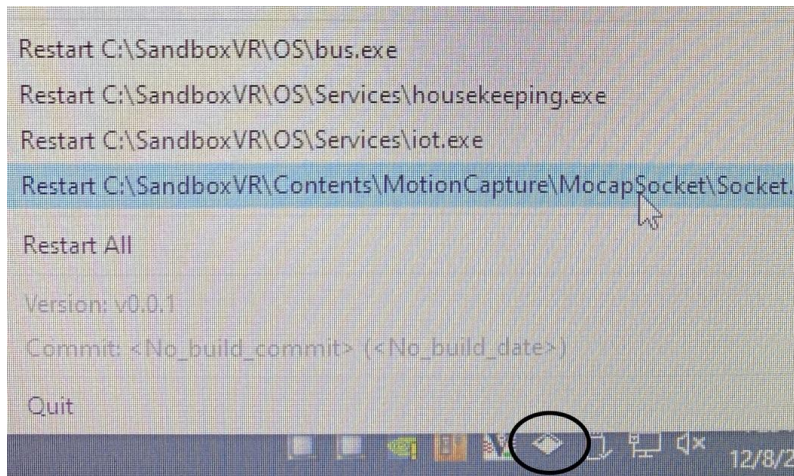
- Double check ALL trackers on guests, make sure trackballs are not covered by clothing, OR no reflective materials on guests.
- Check all equipment to see if there are banded pins or missing tracker balls
- Restart MocapSocket by right click SandboxVROS



- If the problem persists, Slack for assistance as well as notify leads and managers.

H. Players Inside Floor In Experience:

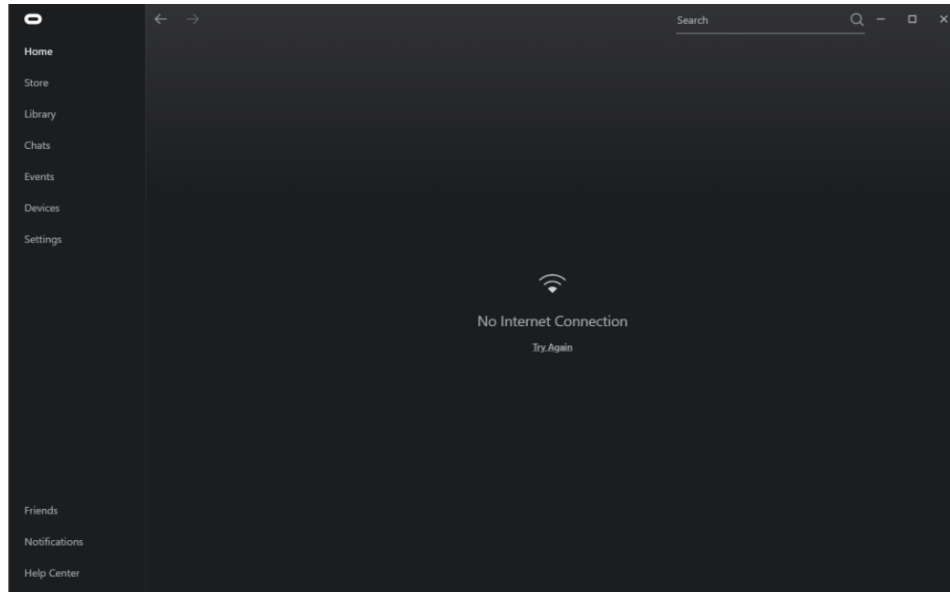
- Restart MocapSocket by right click SandboxVROS (in MOCAP Server)



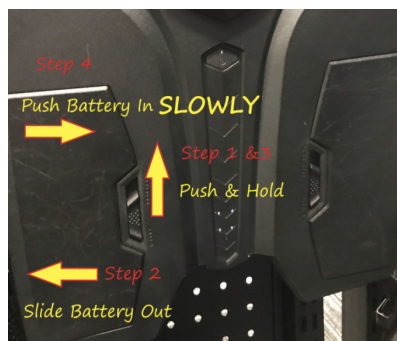
- Ensure that the floor has been properly established. If not, repeat steps 5 and 6 of the calibration process. **(If this occurs during a session, escort them in full gear outside of the room. As this is a fast fix, the players should be able to immediately continue to play once the origin has been established)**
- If the issue persists, right click “sandbox os” and select “restart all”.

If player models are still in the ground after all measures have been taken, report the issue on Slack to get immediate assistance

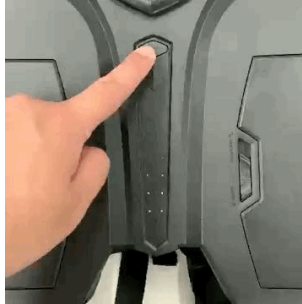
I. PC/BP Disconnection:



- Pause the experience, wait for 30 seconds, most of the time issue resolves itself
- Check backpack battery level, change batteries if battery level is lower than 25%



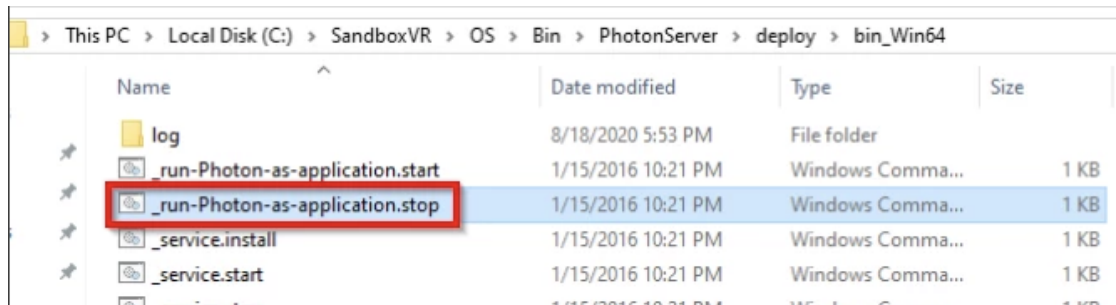
- Launch VNC (shortcuts located on desktop of experience server) to restart SandboxOS
- Power cycle backpack by holding the power button for 7 seconds until LED light goes out, then push it again to restart.
 - **IF** Backpack will not reboot: walk player to gear rack, plug in power cable, power on PC - *sometimes backpacks require a hard cable jumpstart*



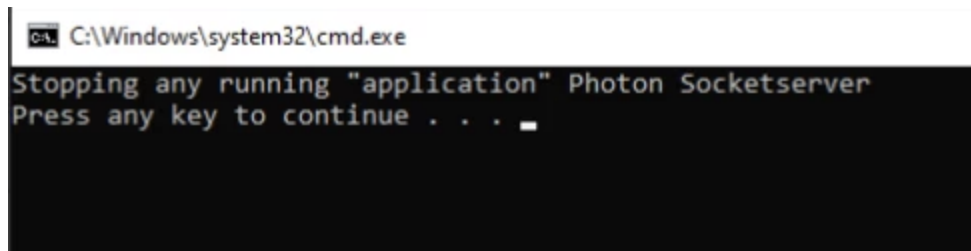
- If problem remains, Slack for assistance as well as notify leads and managers

J. Restart Photon:

1. Open the folder: [C:\SandboxVR\OS\Bin\PhotonServer\deploy\bin_Win64](#)
2. Double click the following file first to stop any running Photon:
[_run-Photon-as-application.stop.cmd](#) (press any key to close the window)

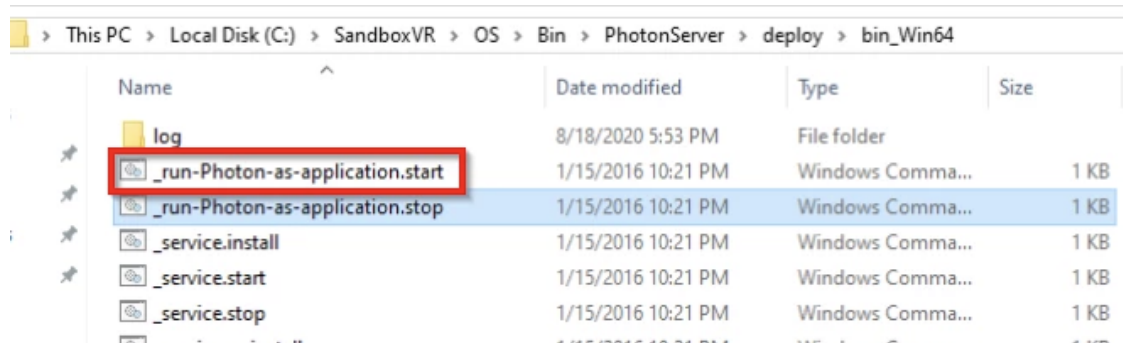


3. Press [Space] to quit the prompt.



***Be very careful not to delete any data during this process. If information is accidentally deleted, contact slack immediately**

4. Then run this file to start Photon: [_run-Photon-as-application.start.cmd](#) (press any key to close the window)



K. Rifles/Pistols Trigger “Stuck”

-Ensure that the issue is a “stuck” weapon by pressing the windows key on the computer. If a stream of “k’s” populate, then you have a “stuck” rifle or pistol.

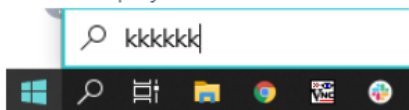
-For a quick fix, right-click “Sandbox Os” and select “Restart All”. In some cases, this will solve the issue immediately.

How to pair a Rifle with dongle

1. Plug the USB dongle into the server.
2. Plug the battery into the gun.
3. Turn on the battery.
4. **Hold** the **Lock** button and keep it held.



5. While holding, turn on the gun by pressing the power button on the gun battery.
6. There should be a light at the front of the gun, flashing rapidly.
7. While **holding** the **Lock** button, the flashing should eventually slow down and become a solid light.
8. Press the Windows key on the server to pull up the Start Menu. Press the trigger of the gun, and the 'K' should display over the Start Menu.



-If the quick fix doesn't work, pull the dongles out one by one to find out which of the equipment is causing the “k's”.

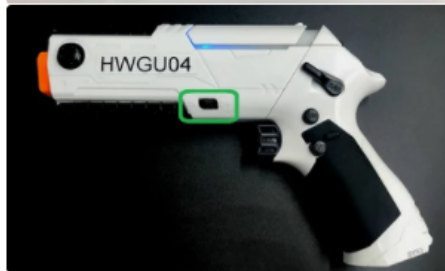
-Once the “k's” stop populating and the correct rifle/pistol has been identified, power cycle the prop, and physically check the equipment to ensure that the buttons are not stuck.

-Re-pair rifle/pistol

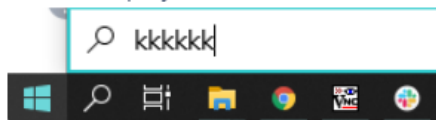
L. Pistol Pairing

How to pair a Pistol with dongle

1. Make sure the Pistol is **switched off** and USB receiver is **removed**. (please note any previous pistol previously synced to the USB receiver, must be switched turned off.)
2. Insert USB dongle into the server.
3. Then **Hold** the "**EN**" button whilst you're turning the On button to the **middle**.



4. Whilst **holding** the "**EN**" down, you will see light flashing **rapidly** below the Pistol rail, this will gradually slow down and becomes a solid, after 10-15 seconds.
5. Press the Windows key on the server to pull up the Start Menu. Press the trigger of the gun, and the 'K' should display over the Start Menu.



M.Gun Pairing

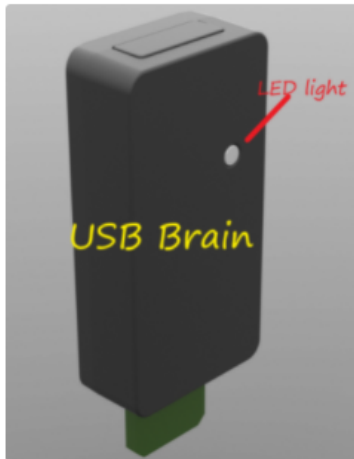
N. Wand Pairing

How to pair a Wand with dongle

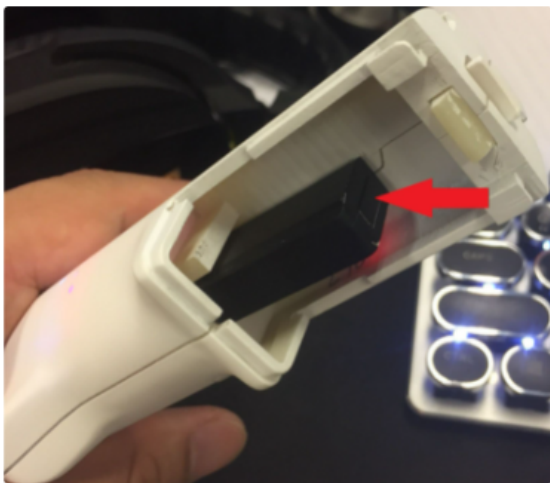
1. Before you plug the Dongle into PC/USB hub, the Brain's LED light should be flashing slowly, which means it has not found the Dongle or it has not been paired.



-
2. Press the button on the **top side** of the Brain, which will make the red LED flash rapidly (Pairing Mode).



3. Plug the Dongle into PC/USB Hub, and press the button on the top of the dongle. Its red LED will also flash rapidly (Pairing Mode).

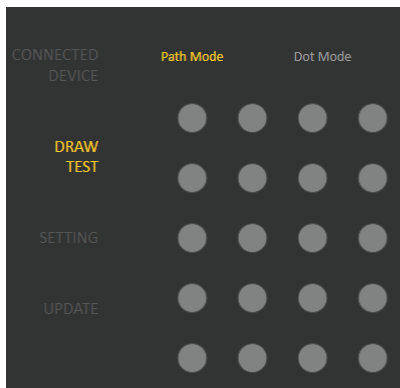
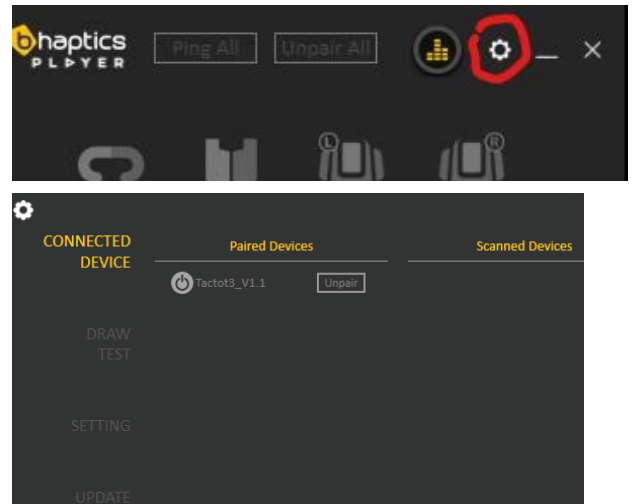


4. Wait for a couple of seconds, both sides' (Dongle & Brain) **BLUE** LED will show Solid light on, which means that they are paired.

O. Haptic Vests: Pairing & Issues

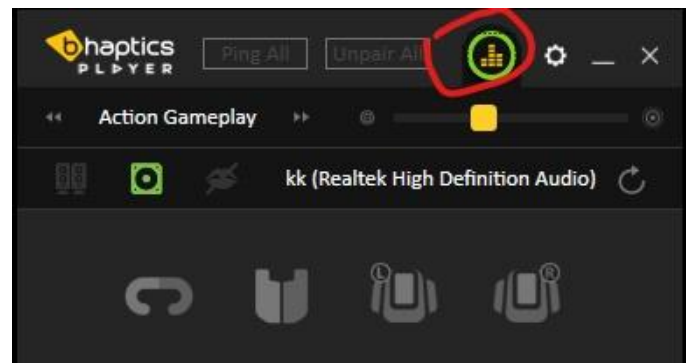
a. Haptic Vest Pairing

- Haptic Vest Indicators
 - **Blue** blinking = attempting to pair bluetooth
 - **Yellow** solid light = vest paired
- Turn on vest so blue light is blinking
- Remote into backpack using VNC
- *Haptics Player* select settings (gear icon)
- Go to *Connected Device*
- Select *Pair* in the *Scanned Device* column
- Test by selecting *Draw Test* and testing haptic vest through haptic feedback to ensure the correct vest is paired



b. Haptics Vest Won't Stop Vibrating

- Remote into backpack using VNC
- *Haptics Player* de-select *Audio To Haptic* icon
- Test
- Reboot if still vibrating
- Replace vest if solution does not work



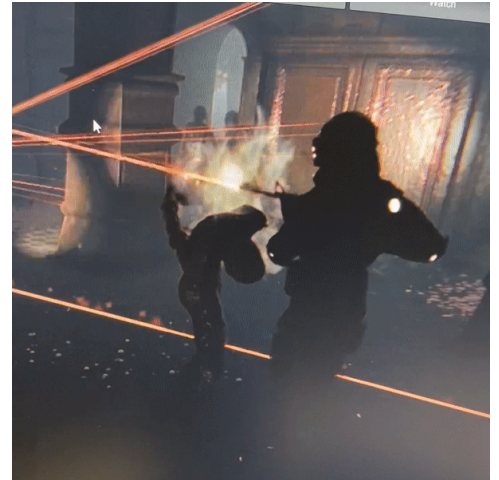
P. In Game Errors

a. HUD Disappears

- Happens occasionally after recording, HUD does not reappear
- Fix: SHIFT + SPACE + T

b. Immortal Zombie

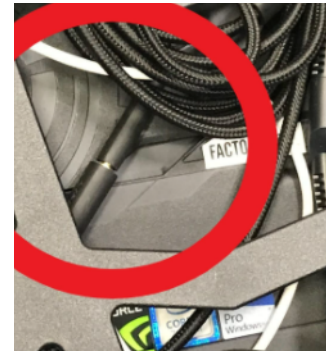
- Deadwood only
- Zombie model will glitch out and not register damage
- Fix:
 - Select *Command*
 - Select *Kill All Enemies*
 - Report Error



Q. Audio Errors

a. Players hear static in headphones

- Check audio cables & headphones (and audio connector)
- Reboot server teamspeak
- Slack it out



b. Players Cant Hear Any Audio

- Check cable connections
- Check audio mixer is ON and not muted
- VNC into PC, check audio settings
- Slack it out

c. Players Cant Hear Each Other

- Reboot server teamspeak & individual player teamspeak
- Slack it out

d. Players Cant Hear Experience

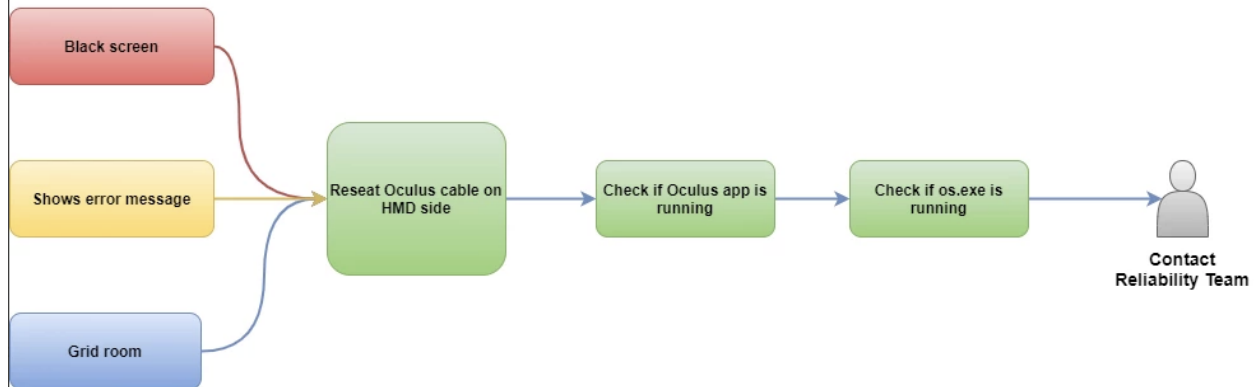
- Reboot server teamspeak
- Slack it out

e. Players Cant Hear Server Mic

- Is the mic plugged in & ON
 - o Front / back of server
- Reboot teamspeak
- Slack it out

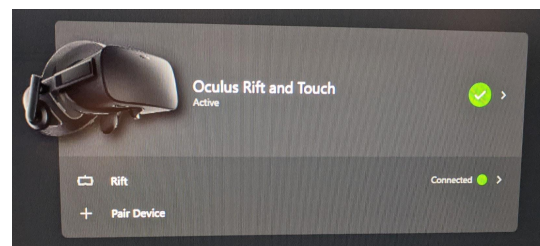
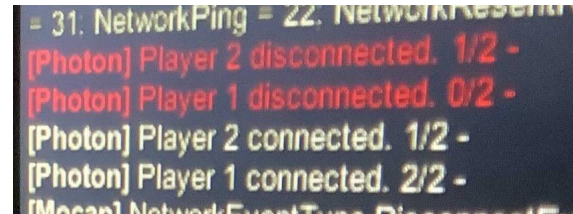
R. Visual Errors

Visual Troubleshooting



a. Oculus Black Screen:

- Pause Experience
- Check Experience
 - Has Photon disconnected?
 - Yes: Will auto-reconnect in 5-10 seconds
- VNC into backpack
 - Is the windows login screen showing?
 - Yes: Login
 - Oculus Launcher
 - Is the connection green?
 - No: Check Oculus cables on both ends (Oculus end HDMI and backpack end HDMI & USB)



- If problem remains, Slack for assistance as well as notify leads and managers

b. Oculus Origin Issue:

VNC into player's backpack, restart Oculus

- If problem remains, Slack for assistance as well as notify leads and managers

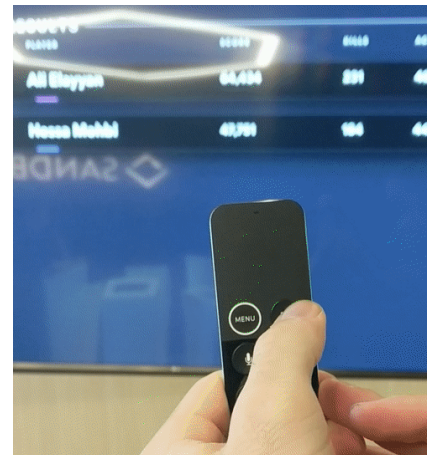
c. Oculus Showing Error Message:

- VNC into player's backpack and restart Sandbox OS/Oculus/Windows 10
- Ensure that the oculus is on (green lit) and that there is internet connection in the oculus
- **Symptom:** "you have a pending hardware issue that is preventing you from starting a VR app" error, Slack for assistance as well as notify leads and managers
- **Symptom:** "unknown source" error, VNC into backpack, go to Oculus *Settings*, select *General*, select ON for *Unknown Source* option
- If problem remains, Slack for assistance as well as notify leads and manager

3. Video & Recap

a. Apple TV not responding

- Go to Barracks Control, select purple *Streaming* Icon
 - Select *Devices*
 - Select *Stop Playing*
 - If not responsive refresh browser, try again
- Apple Remote, reboot Sandbox App (see gif)
- Slack for assistance



b. Rendering Related Issues:

- Select *Rerender*
- If possible, restart server 3 & rerender again
- Slack with session ID, notify guests

c. In Real Life (IRL) Footage Not Recorded

- Footage not recoverable
- Slack with RM # & session ID

d. Rebooting & Resetting Recap

- Do this when:
 - App has frozen, glitched, or is otherwise not responding normally
- Grab AppleTV remote
- Double click TV Icon: 
- Scroll up on touch pad to close out SandboxVR app and all other running apps
- Select menu, and restart SandboxVR app
- Slack out for assistance, notify manager/lead



e. Control Center (Barracks Control)

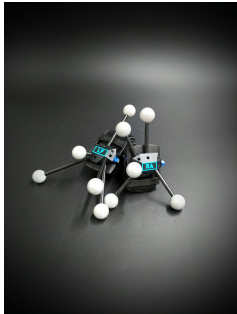
- Must login to the Control Center through @sandboxvr.com email.
- @voice.sandboxvr.com, and other emails will not work

Booking not showing up on Barracks screen

- Is the Booking in the "Room" column?
 - Yes: Move into barracks column, see if that works
 - Booking will automatically skip "Barrack" column when reservation is started in room

4. Equipment

A. Trackers



Sometimes, a tracker may be very obviously damaged. But oftentimes, it's almost impossible to tell the real "condition" of the trackers with the human eye. So sometimes a tracker may look really beat up, but the system will read it just fine, whereas another tracker may look perfect but the system may fail to read it.

Always use the Evoke software to check the tracker's condition. If you are unsure, you can put the trackers inside the experience area and ask the Reliability team to help you check.

This section will teach you how to build a tracker from scratch. The example used below is for the head tracker, but the process is the same for all other trackers, with the only difference being that head trackers use the hex pins (plastic) but the others only use metal.

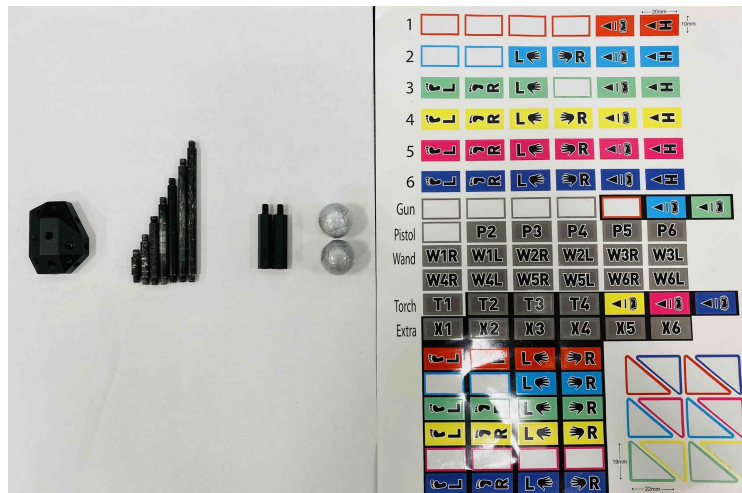
How to build a tracker - head tracker

Video guide:

<https://youtu.be/Bo2TFmXoMyE>

Required materials:

- Tracker Base
- Metal Pins
- Hex Pins (plastic - for head trackers only)
- Tracker ball (markers)
- Glue
- Stickers



- Pliers (optional)

Steps:

Protip: put the **sticker** on the **Tracker Base** first so it's easy to tell what tracker you're working with.

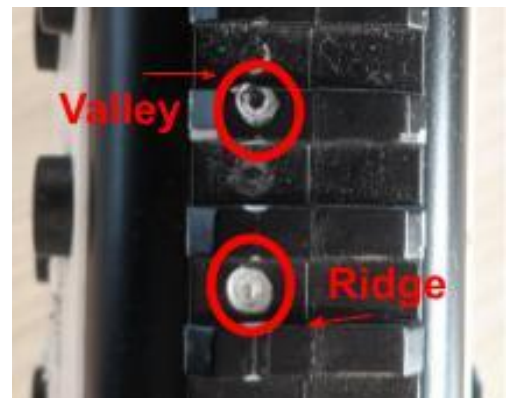
1. Print out or display the [Rigidbody config](#) page on a screen. Locate the particular type of tracker you are building (e.g. Player 6's head tracker in this example)
2. Source the pins necessary (1 10mm and 4 20mm **metal pins**, plus 6 **hex pins** (N) in this case). You can assemble them first (e.g. 1 20mm **metal pin** + 2 **hex pins** for the first pin on the list). Don't assemble the **tracker balls** at this time if the tracker is not going to be used immediately, since they can easily get scratched.
3. Place your **Tracker Base** in the same direction as the picture on the [Rigidbody config](#) so you can locate the holes more easily.
4. Locate the hole and drip one tiny drop of **glue** into it. It helps to double check the holes, but if you mess up, don't worry. Just use **pliers** and a little bit of elbow grease to unscrew the pin and try again. Don't overdo it as it'll spill out and make a mess 😞
5. Screw in the pin before the glue dries. Make sure it's the **metal pin** that threads into the tracker base, not the hex pin (plastic). This is a stress point and if the plastic pin were to break off, we'd have to replace the whole base ☹️
6. Do the same for all pins. Let it rest a few minutes to dry the glue. And you're done! All trackers should have 5 pins each.

How to build all trackers (Version 3.0):

[Rigidbody config](#)

Tracker DON'Ts: Gun & Pistol Tracker Base

- Secure tracker base to RIDGE not VALLEY
 - **Why?**
 - If secured to valley, allen screw may become stuck, and will require drill to remove



With use, the tracker balls or pins may get bent and affect the tracking quality.

B. Replacing Damaged Equipment

a. Damaged Trackers & Pins

Required materials:

- Tracker
- Pliers
- Metal Pins
- Hex Pins (plastic - for head trackers only)
- Tracker ball (markers)
- Glue or Thread Locker

Video guide:

<https://youtu.be/lkESKpAN2ok>

Steps:

1. Unscrew the damaged tracker ball or tracker pins. Note the location (you can refer to the [Rigidbody config](#), too). The pin may be difficult to remove because of the glue. Use the pliers to grab the metal pin and you will be able to unscrew it.
2. Build the replacement pin to the same length.
3. Put a tiny drop of glue into the hole.
4. Screw the pin in. For best practice, check if the tracker can be tracked well by the mocap software. Run a quick test session with it, move around to different parts of the room and make sure the tracking is consistent (no teleporting, stretched limbs, head underground etc issues caused by lost tracking).

C. Regular Maintenance Check

- Battery pin & charger copper pin check
- Wire protector check
- Tracker Pins
 - Not bent
 - ALL pins accounted for
- Tape

- Pistol Hinge
- Wand Tracking Base
- Gun PC/MD Switch

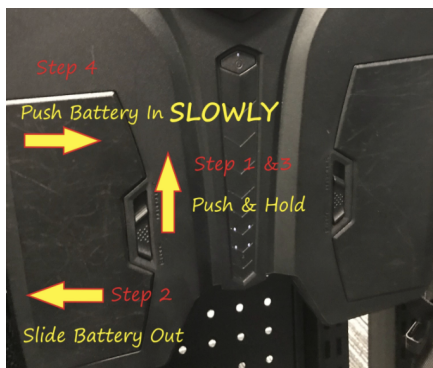
D. Battery To-Know

a. MSI

Battery Levels

- 4 Lights Full = Playable Experience
- 3 Lights Full, 1 Blinking = Playable Experience
- 2 Lights Full, 1 Blinking = **Non-Playable Experience**

Battery Lock



To Replace Battery

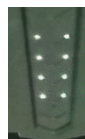
- Lock must be pushed up and held to pull out battery
- Lock must be pushed up and held to replace battery
- IF this is NOT done, the holodeck will have an epidemic of bent copper pins



PC Killer

A battery which will not supply power to a PC, but will sometimes disguise itself as giving power to that PC. Will force other batteries to use all its charge, and will cause PC to run out of power ½ way through experience.

- PC Killer Type 1:
 - Battery will flash with all 4 lights on
- PC Killer Type 2:
 - Battery pin is fused. Will act and look as a fully functional battery.
- Testing For PC Killer
 - Have 2 batteries fully charged in PC



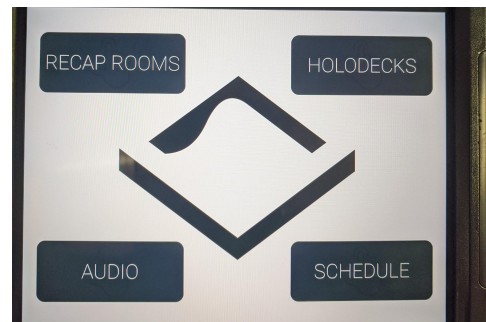
- Turn on PC and unplug.
- Unplug 1 battery.
 - If PC dies, then the other battery is a PC Killer
 - If PC stays on, the other battery is good
- Repeat

5. Qsys and Store Systems Info

SF Pop - MarketST.

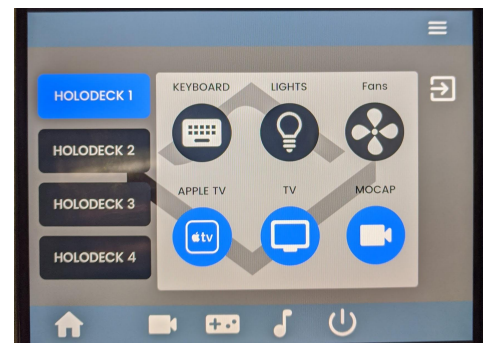
- QSys runs from an app on each tablet, and an independent tablet on the wall in the break room.
- Each app has a home page, where you can select controls for Audio, Recap, Holodecks, and power cycling Schedule.

- Home Screen



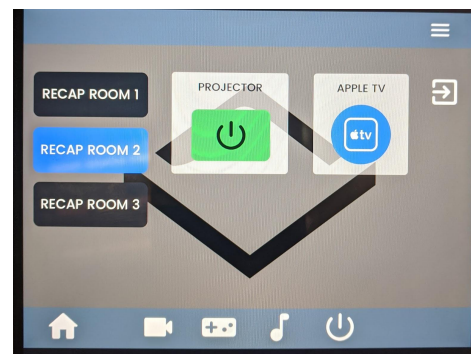
- Holodeck Controls

- Room fan control
- Room lighting control
- Room keyboard light control
- Room TV control
- Room MOCAP control
- Room AppleTV Control
 - Do not turn off AppleTV unless troubleshooting

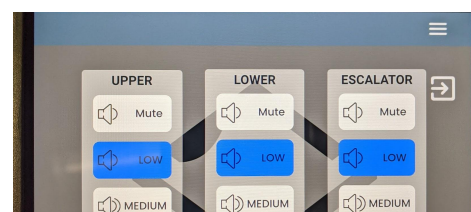


- Recap Controls

- Recap Projector ON/Off
- Recap AppleTV
 - Do not turn off AppleTV unless troubleshooting

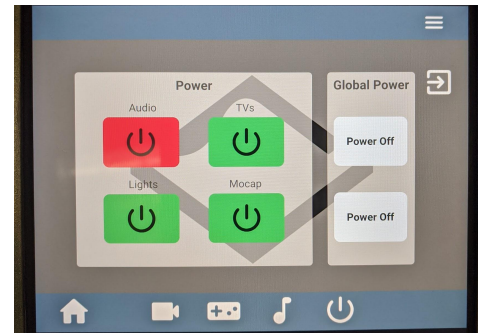


- Sound Controls



- Upper
 - Medium
 - Mix with Sonos controls
- Lower
 - Low
- Escalator
 - N/A

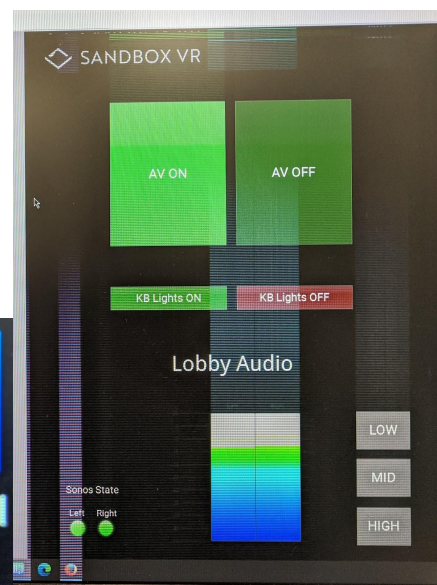
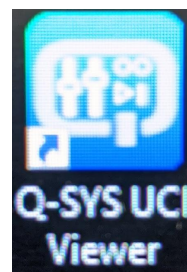
- Power Controls
 - Recommend do not touch
 - Power
 - On/Off for indicated systems
 - Global Power
 - On/Off



Escalator Lighting	192.168.45.91 192.168.45.92 192.168.45.93	● Select <i>Settings</i> ● Go to to bottom of page to <i>Fader Test</i> ● Select <i>Enable Fader</i> ● Move the B tab to ~76
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SanMateo - Hillsdale

- Audio Visual Controls
 - AV ON
 - Turn on SONOS
 - Turn on main store power
 - AV OFF
 - Turn on SONOS



- Turn on main store power
- Location: on main server rack, under *Qsys UCI Viewer*

- Lighting & Main Store Controls
 - IP: 192.168.4.241
 - Login: *info location tbd*
 -
- Security Cameras
 - IP: 192.168.4.240
 - Login: *Info location tbd*

-pic/breakdown of main lights & system (put in IP table as well, with quicklink to to this section)

-pic/breakdown of Qsys for lights/music

Electrical Room
1 light/device 1 control
Experience Room A
9 lights/devices 6 controls
Experience Room B
7 lights/devices 4 controls
Experience Room C
8 lights/devices 6 controls
Experience Room D
7 lights/devices 4 controls
Reception
6 lights/devices 2 controls
Restroom
1 light/device 3 controls
Storage
1 light/device 1 control
Waiting Area
5 lights/devices 1 control

Location 3

Location 4

Location 5

IP Table

Use	IP	Notes
Room Control Panel <i>Room 4 example</i>	10. 4 .1.1	The 2nd digit indicates which room you are in <i>Room 1 = 1</i>

<i>Use Browser NOT VNC</i>		<i>Room 2 = 2 Room 3 = 3 Room 4 = 4</i>
Remote into Server VNC	10.4. 1 ._	The 3rd digit = 1 indicates you are remoting into a server
Remote into Server VNC	10.4.1. 3	The 4th digit = 1-3 indicates which server in the server rack you are remoting into <i>Experience = 1 Server = 2 Utility = 3</i>
Remote into PC's VNC	10.4. 2 ._	The 3rd digit = 2 indicates you are remoting into a PC
Remote into PC's VNC <i>PC 6 example</i>	10.4.2. 6	The 4th digit = 1-8(or other) indicates which PC you are remoting into <i>PC 1 = 1 PC 2 = 2 PC 3 = 3 etc...</i>

For Sonos Related Issue:

- Slack

In-Store WIFI Issues:

- Slack

Apple TV Related Issues:

- Slack

6. Printable Guide

Emergency Troubleshooting Guide

Rules of Troubleshooting:

1. When in doubt, pause experience
2. Check the hardware
3. If you don't know the answer, slack it and call management
4. Finagle's Law: *"Anything that can go wrong, will - at the worst possible moment."*

→ *Symptom*

◆ *Resolution*

→ Headset black & headphones no audio - *start of experience*

- ◆ Check cables
- ◆ VNC into BP, log into Windows
- ◆ Slack/call Management if not fixed

→ Players stuck in the floor

- ◆ Switch to Server 2, right click Sandbox icon in taskbar, *Restart All*
- ◆ Disable and Re-enable *Strobes*
- ◆ Slack / call management if not fixed

→ Player avatar is shorter, or taller than they should be

- ◆ Select *Command* (in experience)
- ◆ Select *Reset Calibration*
- ◆ Slack / call management if not fixed

→ Player disconnected in experience - *Unity text: [Photon] Player 1 disconnected.*

- ◆ Wait for 10-30 seconds for photon reconnection
- ◆ Slack / call management if not fixed

→ Player disconnected in experience OR headset goes black- *No unity text*

- ◆ Is the computer off, asleep, or needs new batteries
- ◆ Check cabling and connections
- ◆ Slack / call management if not fixed

→

Experience Shortcuts

(during experience on Server 1)

Function	Keys	Notes
Toggle full-screen view	Alt + Enter	No button available on Console
Toggle Console	Shift + Space + T	
Toggle Log	Shift + Space + L	
Start Game	Shift + Space + S	
Pause / Resume Game	Shift + Space + P	
Force Defeat	Shift + Space + E	
Go to Wave	Shift + Space + Alpha #	Select the number of the wave from the alphanumeric row
Kill Player	Shift + Space + Keypad #	Select the number of the player from the keypad
Solve OilBarrelPuzzle	Shift + Space + Q	Only available for Davy Jones

Calibrating Cameras:

(To be done every morning before room is in use and at manager's discretion during operating hours)

1. ENABLE Strobes
2. "Start Masking" for 20 seconds, then "Stop Masking"
3. DISABLE Strobes
4. "Start wave in camera" and waving the T-wand in front of each camera. (cameras need to be under 0.1 or very close to that range and in green). *If you see image error is greater than 0.175(see right), repeat step 4.*
6. ENABLE Strobes
7. Place the T-wand on the center floor (look for the cross) with the long end pointing at the fans, then turn on the wand
8. "Start Set Origin" for 10 seconds, then "Complete set origin"

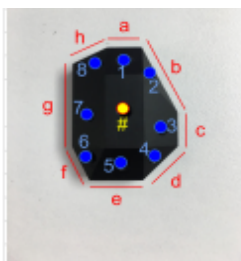
Vicon Motion Capture System Calibration Guide



100%		
Camera	Wand Count	Image Error
1 Vero v2.2	3150	4.538
2 Vero v2.2	1405	6.611
3 Vero v2.2	1013	3.259
4 Vero v2.2	4371	2.436
5 Vero v2.2	4436	1.563
6 Vero v2.2	8080	1.567
7 Vero v2.2	1994	1.512
8 Vero v2.2	2147	1.630
9 Vero v2.2	1000	1.960
10 Vero v2.2	1316	1.590
11 Vero v2.2	3194	1.852
12 Vero v2.2	4949	4.311

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Change:
redesign track for optitrack supporting PVP tracking

